

CHE 396 Quiz 1

William Rawson

TOTAL POINTS

95 / 100

QUESTION 1

11 15 / 15

- **3 pts** no detail explanation about the process
- **1 pts** no applications mentioned
- ✓ - **0 pts** correct
- **15 pts** incorrect

QUESTION 2

2 2 15 / 15

- **12 pts** other processes not mentioned
- **3 pts** no brief information about the processes
- ✓ - **0 pts** correct

QUESTION 3

3 3 15 / 15

- **10 pts** Major logical error in the BFD
- **5 pts** Minor mistake in the BFD
- ✓ - **0 pts** correct

QUESTION 4

4 4 15 / 15

- **15 pts** separation process not mentioned
- **10 pts** Partial process
- **5 pts** Minor error
- ✓ - **0 pts** correct

QUESTION 5

5 5 5 / 5

- **2.5 pts** partial reaction
- ✓ - **0 pts** correct
- **5 pts** incorrect

QUESTION 6

6 6 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** wrong technique

- **2.5 pts** partially correct

QUESTION 7

7 7 5 / 5

- **5 pts** No full form or production process
- **3.5 pts** full form but no production process
- ✓ - **0 pts** correct
- **2 pts** partially production process
- **1.5 pts** no full form

QUESTION 8

8 8 5 / 5

- ✓ - **0 pts** Correct
- **2 pts** Partially correct
- **4 pts** wrong
- **5 pts** not written

QUESTION 9

9 9 15 / 20

- **10 pts** incorrect logic for recovery rate (step 1)
- **10 pts** incorrect logic for recovery rate (step 2)
- **0 pts** correct
- **5 pts** correct logic but calculation error for recovery rate (step 1)
- ✓ - **5 pts** correct logic but calculation error for recovery rate (step 2)

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Name

1. What is the design basis for your mini-project?

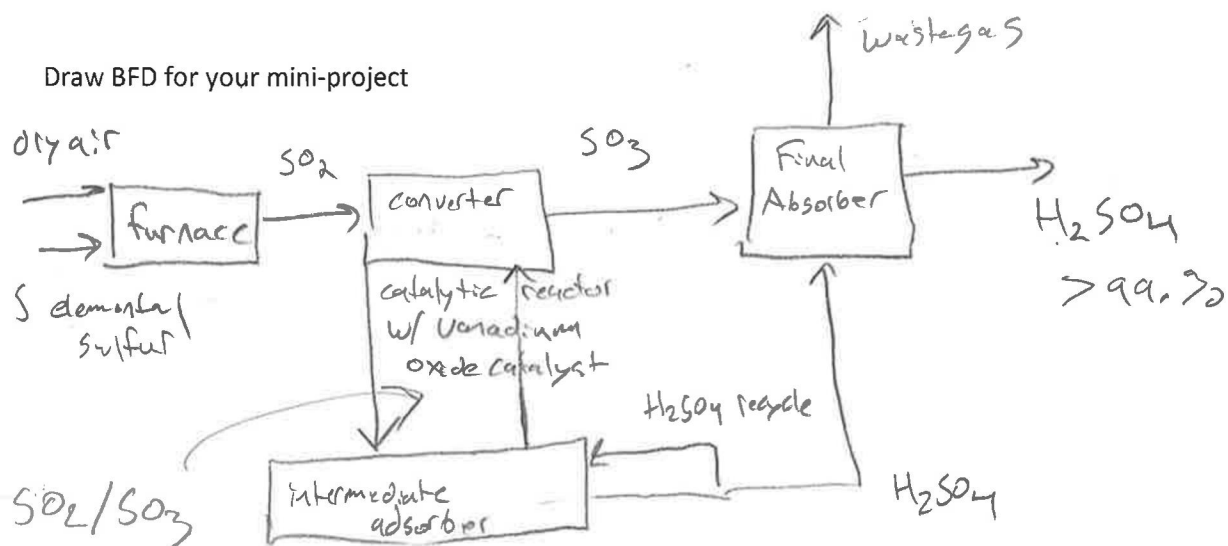
The design basis was to make 300,000 ^{ton}/_{year}.

The location was set in the Midwest and the units for our process were in the English system.

2. List competing processes.

Competing processes were single contact absorption and WSA process.

3. Draw BFD for your mini-project

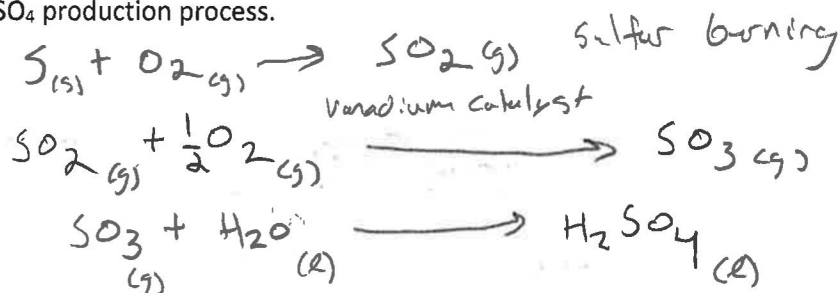


4. Explain separation section of your mini-project in details.

Elemental Sulfur is burned to produce ~1070 volume SO_2 gas, this sulfur is burned with dry air. This SO_2 gas is then put into a catalytic reactor and is reacted to produce SO_3 . The reactor for the double contact method has 4 stages and is done using a Vanadium oxide catalyst. At the 3rd stage the SO_3 converted is removed and sent to an intermediate adsorber. This takes advantage of Le Chatelier's Principle and you get better conversion of SO_2 to SO_3 . The SO_3 leaves the 4th stage of converter to go into the Final absorber. Both absorbers mix SO_3 gas with concentrated H_2SO_4 to create oleum, H_2SO_7 , which is then diluted down with water to produce >99.3% pure H_2SO_4 .

5. Write down the chemistry of H_2SO_4 production process.

implied in 3rd rxn is
oleum production.



6. Explain how paraxylene is separated from other xylenes (ortho and meta)?

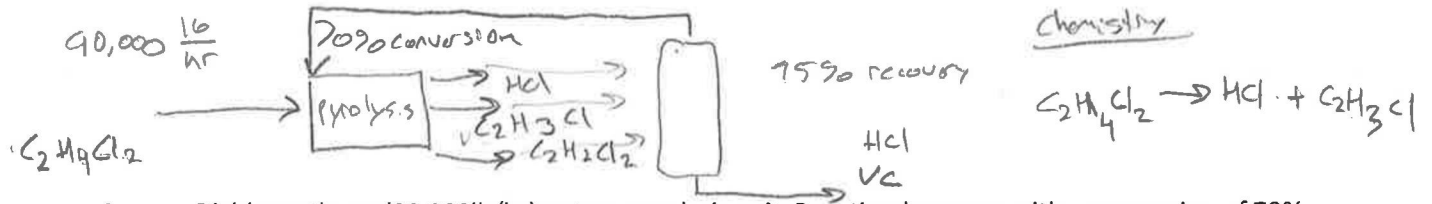
Paraxylene is separated out from other xylenes by a specially designed particle that adsorbs paraxylene. B/c paraxylene is longer () than the other xylenes, it gets stuck in the particle while the others pass through. This is done as the Parex method.

7. What is LNG? Explain briefly its production.

LNG is liquified natural gas. It can be taken from shale deposits and goes through separations to make it pure. No reactions take place.

8. How oxygen is produced in chemical industry?

Oxygen is produced in chemical industry by cryogenic separation.



9. Dichloroethane (90,000 lb/hr) enters pyrolysis unit. Reaction happens with a conversion of 70% producing vinyl chloride and HCl. After reaction dichloroethane is separated by distillation with a recovery rate of 95% and recycled back to the reactor. Calculate the vinyl chloride output flow rate and recycle stream flow rate in lb/hr.

$$\text{Conversion} = \frac{\text{amount A consumed}}{\text{amount A supplied}} \Rightarrow 0.70 = \frac{\text{amount A consumed}}{909.09 \frac{\text{lbmol}}{\text{hr}}}$$

Dichloroethane $\text{C}_2\text{H}_4\text{Cl}_2 = 99$

Vinyl chloride $\text{C}_2\text{H}_3\text{Cl} = 62.5$

HCl = 36.5

636.36 $\text{C}_2\text{H}_4\text{Cl}_2$ consumed

$\therefore 272.72 \frac{\text{lbmol}}{\text{hr}}$ left

$$900,000 \frac{\text{lb}}{\text{hr}} / 99 \frac{\text{lb}}{\text{lbmol}} = 909.09 \frac{\text{lbmol}}{\text{hr}} \text{ Supplied}$$

1:1 mol ratios

Vinyl chloride output

Overall system balance.

$909.09 \frac{\text{lbmol}}{\text{hr}}$ comes in, $909.09 \frac{\text{lbmol}}{\text{hr}}$ goes out.

$$\therefore 909.09 \frac{\text{lbmol}}{\text{hr}} \cdot 62.5 \frac{\text{lb}}{\text{lbmol}} = 56,818.18 \frac{\text{lb}}{\text{hr}} \text{ w/ 1:1 mol ratio of VC to } \text{C}_2\text{H}_4\text{Cl}_2$$

Output of vinyl chloride is $56,818.18 \frac{\text{lb}}{\text{hr}}$

If 95% recovery of $272.72 \frac{\text{lbmol}}{\text{hr}}$ flowrate of $\text{C}_2\text{H}_4\text{Cl}_2$,
that's $259.08 \frac{\text{lbmol}}{\text{hr}}$ recovered.

$$259.08 \frac{\text{lbmol}}{\text{hr}} \cdot 99 \frac{\text{lb}}{\text{lbmol}} = 25,649.32 \frac{\text{lb}}{\text{hr}} \text{ } \text{C}_2\text{H}_4\text{Cl}_2 \text{ recycled.}$$